

Probabilistic Prediction of Placements in Teamfight Tactics:

A Bayesian Nonparametric Approach with BART

Extended Abstract

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Problem. In Teamfight Tactics (TFT), eight players compete until their health reaches zero, so a match maps naturally onto a survival-analysis problem in which event time is the final round a player reaches. This work predicts each player’s dynamic elimination probability at every round, using *only* information available *before* the target match: the player’s recent history and how they compare to the lobby. No in-game information from the target match is used as a predictor.

Method. The match is reframed as sequential binary classification over discrete rounds and modeled with **Bayesian Additive Regression Trees (BART)** under a probit link, using Albert–Chib latent-variable data augmentation and MCMC sampling. The full pipeline was built in R: extraction through the **Riot Games API** (PUUID resolution; patches 16.4–16.6), per-player history construction, a 10-match burn-in with empirical-Bayes shrinkage to stabilize metrics, lobby-relative feature engineering (50+ leakage-free covariates), and longitudinal expansion into a discrete survival matrix. Four specifications were benchmarked (full vs. PCA-reduced covariates; partial vs. enriched lobbies) using Integrated Brier Score, Log-loss, Spearman’s ρ , Kendall’s τ , and the Concordance Index (C-index).

Key results.

- On simulated benchmarks (Sparapani et al., 2016), BART systematically outperformed the Kaplan–Meier estimator on RMSE/MAE, including under non-proportional hazards with crossing survival curves.
- On real tournament data, the selected model (`model_full`: original dataset, full covariates) recovered the relative ranking of players within a lobby, maximizing all three concordance estimators (C-index = 0.630, ρ = 0.350, τ = 0.345). A calibration–discrimination trade-off was observed: enriched models produced better-calibrated absolute probabilities but lost ordering ability.
- The estimated survival curves reproduced the staged macro-dynamics of TFT (plateaus on PvE/carousel rounds) and the empirical elimination order of lobbies.
- **All** observed eliminations fell within the model’s 90% and 95% credible intervals; for 20 of 24 participants the predicted median was within 4 rounds of the actual elimination round.

Honest limitation. The credible intervals are wide (often spanning 2+ game stages). Pre-match signal robustly predicts a lobby’s *relative merit ordering* but is insufficient to isolate the *exact* round of elimination: TFT’s in-game variance (shop randomness, matchmaking, immediate tactical decisions) is irreducible from pre-match information alone. This defines a realistic ceiling for any pre-match predictive product and prevents over-promising on pointwise accuracy.

Takeaway. A Bayesian nonparametric ensemble, trained purely on smoothed past behavior, learns non-proportional hazards and recovers *who* is likely to do well — while honestly bounding how precisely *when* a player busts can be known.